

Assignment -2

Name: - Aditya Ingale

Roll no.: - 260250320005

1. Create a medical bot application that can prescribe medication based on the patient's symptoms and age.
Hint:(Your task is to create a C# console application that simulates a medical bot named Bob. Bob should be able to prescribe medication based on the symptoms of a patient passed as an argument.)

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Assignment2
{
    internal class MedicalBot
    {
        public string Name;

        public MedicalBot(String name)
        {
            Name = name;
        }

        public string Prescribe(string symptom, int age)
        {
            string medication;

            if (symptom == "fever")
            {
                medication = (age < 12) ? "Paracetamol (child dose)" : "Paracetamol (adult
dose";
            }
            else if (symptom == "cough")
            {
                medication = (age < 12) ? "Children's cough syrup" : "Cough lozenges";
            }
            else if (symptom == "headache")
            {
                medication = (age < 18) ? "Consult a doctor before giving medication" :
"Ibuprofen";
            }
            else
            {
                medication = "Symptom not recognized - please consult a doctor";
            }

            return $"{Name} prescribes: {medication}";
        }
    }
}

using Assignment2;

Console.WriteLine("Hello, World!");

MedicalBot bob = new MedicalBot("Bob");

Console.WriteLine(bob.Prescribe("fever", 8));
Console.WriteLine(bob.Prescribe("headache", 25));
Console.WriteLine(bob.Prescribe("cough", 30));
Console.WriteLine(bob.Prescribe("rash", 20));
```

```
Hello, World!
Bob prescribes: Paracetamol (child dose)
Bob prescribes: Ibuprofen
Bob prescribes: Cough lozenges
Bob prescribes: Symptom not recognized - please consult a doctor
```

```
D:\Code\C#\Assignment2\bin\Debug\net8.0\Assignment2.exe (process 19348) exited with code 0 (0x0).
Press any key to close this window . . .
```

2. Create simple C# code that converts various values from one type another type.

```
Console.WriteLine("---- Type Conversions ----");

int wholeNumber = 100;
double implicitToDouble = wholeNumber; //
Console.WriteLine("Implicit (int to double): " + implicitToDouble);

double decimalValue = 9.78;
int explicitToInt = (int)decimalValue;
Console.WriteLine("Explicit cast (double to int): " + explicitToInt);

string numberAsString = "245";
int convertedFromString = Convert.ToInt32(numberAsString);
Console.WriteLine("Convert.ToInt32 (string to int): " + convertedFromString);

double convertedToDouble = Convert.ToDouble("78.45");
Console.WriteLine("Convert.ToDouble (string to double): " + convertedToDouble);

string boolAsString = "true";
bool parsedBool = bool.Parse(boolAsString);
Console.WriteLine("bool.Parse (string to bool): " + parsedBool);

string userInput = "abc";
bool success = int.TryParse(userInput, out int result);

if (success)
{
    Console.WriteLine("Parsed successfully: " + result);
}
else
{
    Console.WriteLine("TryParse failed safely - input was not a valid number, no crash occurred");
}
```

```
---- Type Conversions ----
Implicit (int to double): 100
Explicit cast (double to int): 9
Convert.ToInt32 (string to int): 245
Convert.ToDouble (string to double): 78.45
bool.Parse (string to bool): True
TryParse failed safely - input was not a valid number, no crash occurred

D:\Code\C#\Assignment2\bin\Debug\net8.0\Assignment2.exe (process 26152) exited with code 0 (0x0).
Press any key to close this window . . .|
```